# Harpreet Kour

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## PROFESSIONAL SUMMARY

Software Development Engineer pursuing Integrated M.Tech in Artificial Intelligence and Computer Engineering (CGPA: 9.09/10) at VIT Bhopal. Completed 272+ hands-on GCP labs covering Kubernetes, Terraform, VPC networking, and fault-tolerant distributed systems. Built and deployed a live Ethereum validator node monitoring dashboard using Web3.py and Docker. Passionate about high-availability infrastructure, proof-of-stake blockchain networks, and scalable backend systems with 99.99% uptime.

## WORK EXPERIENCE

**Google Cloud Generative AI Intern** Nov *2025 - Jan 2026*

*SmartBridge (SmartInternz) - Virtual | AICTE Approved*

- Earned verified completion certificate (Certificate ID: VIP-AI-2026-30668) for proficiency in Vertex AI and Generative AI APIs; recognized as a Google Cloud Innovator in December 2025.
- Applied GCP ecosystem learnings to build and evaluate real-world GenAI workflows, reinforcing skills validated by 5+ concurrent Google and Cloud certifications.

## PROJECTS

**Ethereum Validator Node Monitor and Dashboard** | Python, Web3.py, Streamlit, Docker *March 2026*

- Developed an Ethereum mainnet monitoring dashboard tracking live block data, gas prices, validator metrics, and network health by connecting to the Ethereum JSON-RPC API via Web3.py.
- Containerized with Docker and Docker Compose for production-grade deployment; published live on Streamlit Cloud with a configurable gas price alert system.

**Hindi AI Assistant** | Python, Streamlit, SpeechRecognition, OpenCV *November 2025*

- Built a full-stack multilingual conversational AI with speech-to-text, rule-based and LLM response generation, text-to-speech, and real-time face detection serving seamless Hindi-language interactions.
- Integrated four modular AI pipelines (ASR, NLU, TTS, CV) into a single Streamlit UI, reducing latency between input and response by architecting asynchronous processing flows.

**SmartHelmet - Alcohol and Drowsiness Detection** | Python, IoT, MQ-series, MPU6050 *Nov - Dec 2025*

- Designed an IoT-based Smart Helmet Safety System detecting alcohol consumption and drowsiness using MQ-series alcohol sensors and MPU6050 motion tracking to enhance rider safety.
- Achieved sub-second alert response times by optimizing sensor polling intervals, demonstrating embedded AI deployment on resource-constrained hardware.

**E. coli Resistance Prediction (Research Paper)** | Python, Scikit-learn, ML *August 2025*

- Developed machine learning models to predict E. coli presence and antibiotic resistance patterns in environmental samples, enabling rapid water quality assessment and public health risk mitigation.
- Paper forthcoming in African Journal of Biomedical Research; annotated and preprocessed environmental datasets achieving significant accuracy improvements in resistance classification.

**MERN Stack Web Application** | MongoDB, Express.js, React.js, Node.js *August 2025*

- Developed a full-stack web application with CRUD operations, RESTful backend architecture, and dynamic front-end integration using the complete MERN stack.

## EDUCATION

**Vellore Institute of Technology Bhopal** 2022 - 2027

*Integrated M.Tech in Artificial Intelligence and Computer Engineering | CGPA: 9.09/10*

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, C++, SQL

**AI and ML:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Vertex AI, Generative AI APIs

**Cloud and DevOps:** Google Cloud Platform, Kubernetes, Terraform, Docker, Jenkins, Linux, VPC Networking, Load Balancing, Cloud SQL, Dataflow

**Blockchain:** Web3.py, Ethereum JSON-RPC, Proof-of-Stake, Validator Node Monitoring, Smart Contracts

**Web and Backend:** Node.js, Express.js, React.js, MongoDB, Django, REST APIs

**Tools:** Git, GitHub, Postman, Jupyter Notebook, Streamlit, Docker Compose

**CS Fundamentals:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Design Patterns

## ACHIEVEMENTS

- Google Cloud Innovator: Recognized as a Google Cloud community innovator (December 2025) after completing 272+ GCP labs with high scores.
- HackRx 4.0 by Bajaj Finserv: Developed Android app for forgery document detection using machine learning.
- NPTEL Cloud Computing: Completed with Elite and Silver rank, demonstrating advanced cloud proficiency.
- Leadership: Volunteered at AI Club - VIT Bhopal organizing workshops and seminars; supported outreach for Mood Indigo - IIT Bombay, India's largest cultural festival.